

Lecture-25

Pole development
venerdì 23 aprile 2021 11:27

Zero value Time constant Method

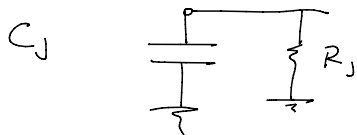
Independent Voltage

and current $\tau_k \gg \tau_1, \tau_2$
generators are nullified

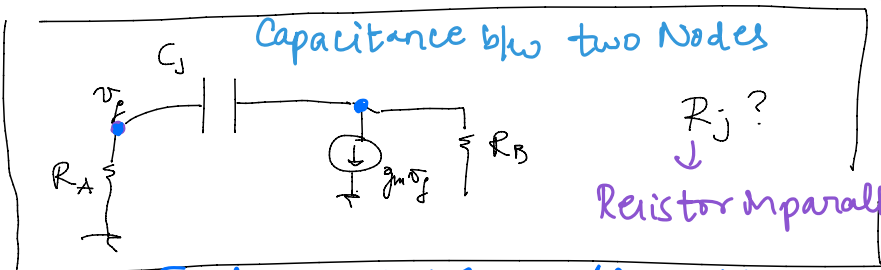
$$\omega_{-3dB} = \frac{1}{\sum \tau_j}$$

$$\tau_j = C_j \cdot R_j$$

R_j Resistance in parallel to the Capacitance



If the capacitance is connected to a Node and GND then it's easy to find the Resistance which is connected in parallel to it.

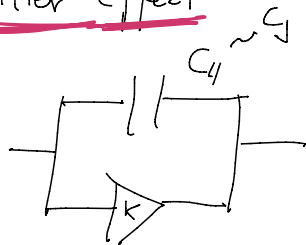


For this kind of configuration we have two options

B)

$$K = -g_m R_B$$

1. Miller effect



$$\equiv \frac{1}{s} C_{||} \cdot (1-K) + \frac{1}{s} C_{||} \left(1 - \frac{1}{K}\right)$$

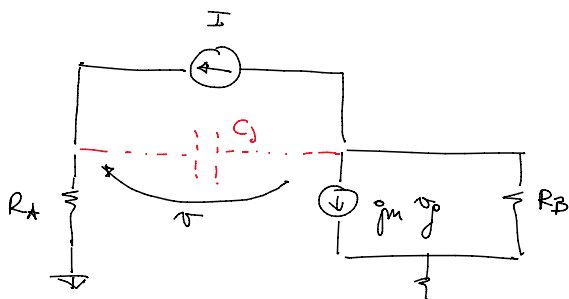
$$C_{JA} = C_{||} (1 + g_m \cdot R_B) \approx C_{||} \cdot g_m R_B$$

$$\tau_A = R_A \cdot C_{JA} \approx R_A \cdot C_{||} \cdot g_m R_B$$

Here $K = \text{Gain} = -g_m R_B$

" In order to calculate gain K capacitance is not considered "

2.



$$\frac{V}{I} = R_{probe} = R_A + R_B + g_m R_B \cdot R_A$$

$$R_{probe} \approx g_m R_B \cdot R_A$$

$$\tau_j = R_{probe} C_j = g_m R_B \cdot R_A \cdot C_j$$

This Technique we studied in Lecture-23

Note:- In most circuits the capacitance which is connected to the most nodes will become the Most Dominant pole.

→ An expert Analog designer just looks at the most relevant capacitance with quick look estimates what are the resistances in parallel to that specific capacitance. which will help in identifying the Dominant pole and thereby Cutoff Frequency.

→ There are lucky cases where the Miller capacitance is negligible because we don't have floating capacitances b/w high gain stages and this is the case for example

a " Folded cascode design."

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Design Specification

SR Slew Rate

ICMR ✓ I/p Common Mode Range

OR ✓ → o/p Range

→ Gain ✓ → DC Gain

f_{-3dB} → Dominant Pole

Output Range

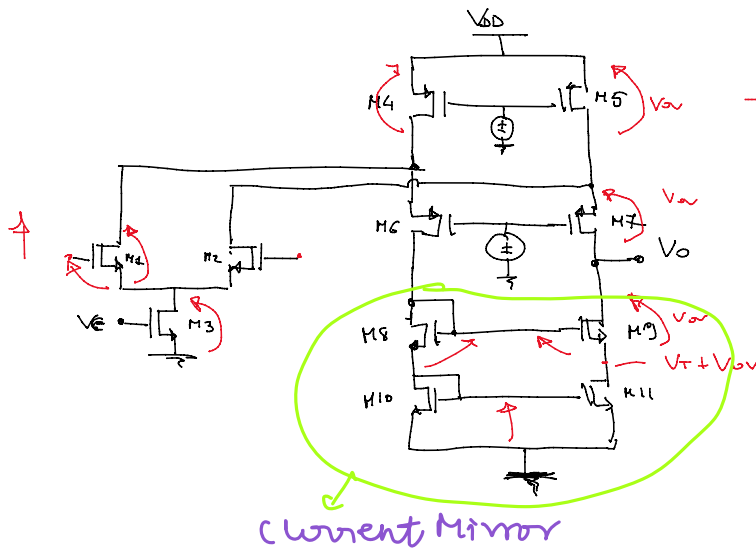
$$V_{OMAX} = V_{DD} - V_{OV5} - V_{OV7}$$

OR $V_{OMIN} = V_T + V_{OV} + V_{OV}$

Input Range

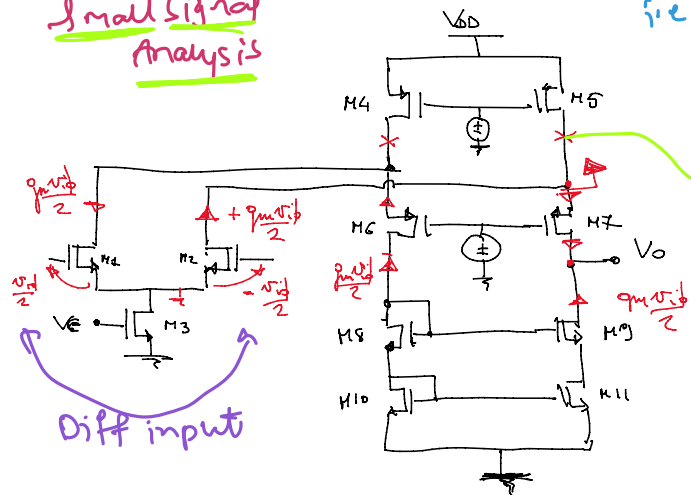
$$MAX = V_{DD} - V_{OV4} - V_{OV2} + V_{OV1} + V_T$$

V_{iCH} MIN = $V_{OV1} + V_T + V_{OV3}$



Differential Amplifier

Small Signal Analysis



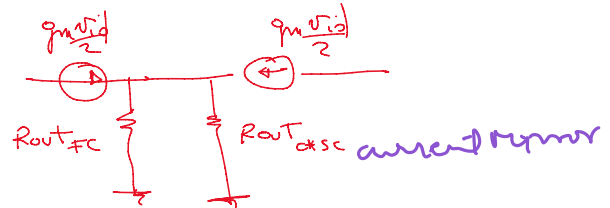
$$R_{out_{cascode}} = g_m \cdot v_o^2$$

$$R_{out_{fc}} = g_m v_{o2} (r_{o5} // r_{o2})$$

current mirror

Intuitive Analysis to calculate DC gain

i.e. $v_{icm} = 0$

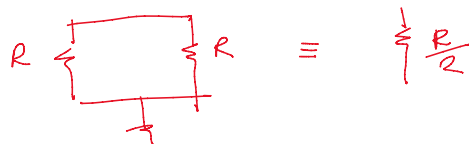


$$R_{out_{fc}} // R_{out_{cascode}} = R_{out}$$

$$v_o = g_m \cdot v_{id} \cdot R_{out}$$

$$\frac{v_o}{v_{id}} = g_m \cdot R_{out}$$

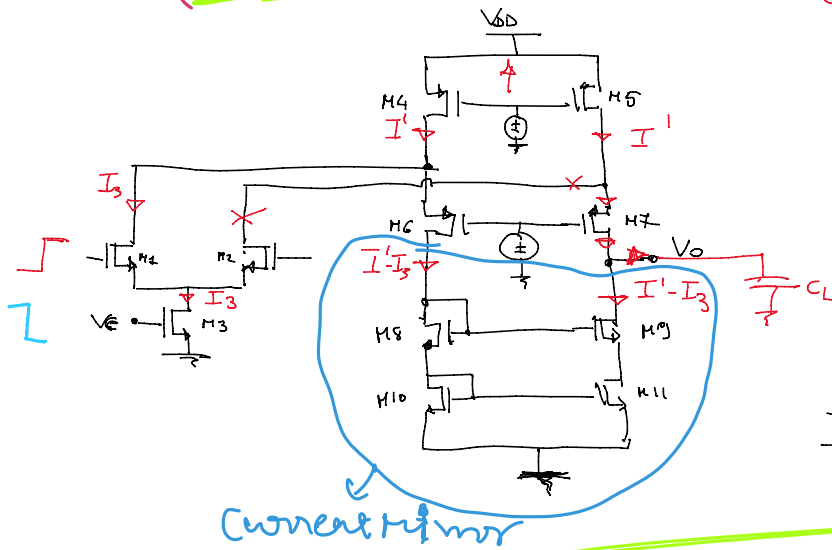
Differential Gain



Similar to analysis we have done in earlier classes

Here we assume that the current flowing through M_4 & M_5 is almost zero because the input conductance of common gate M_6 & M_7 is very high.

Slew Rate (DC Analysis)



$$SR = \max \left| \frac{dV_{out}}{dt} \right|$$

$$I' - (I' - I_3) = I_3$$

$$\max \left| \frac{dV_{out}}{dt} \right| = \frac{I_3}{C_L}$$

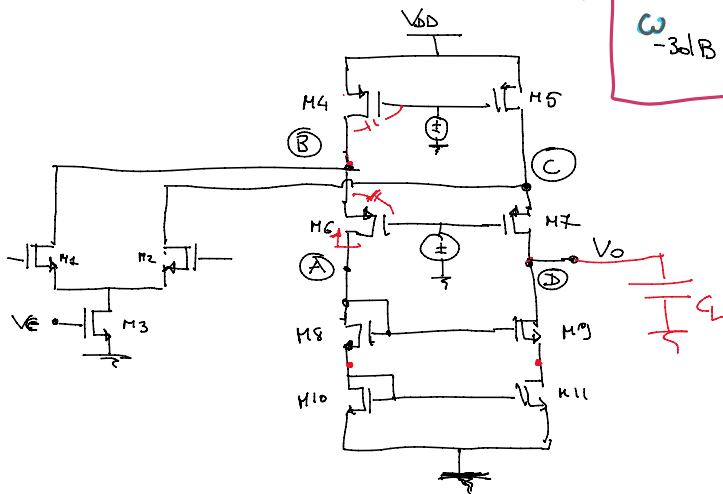
Slew Rate

$$\frac{I_3}{C_L}$$

$$I' > I_3$$

Cutoff frequency

$R_j C_j$: Time constant

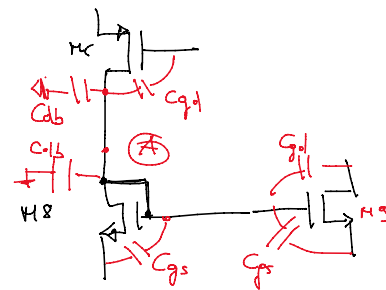


$$\omega_{-3dB} = \frac{1}{\sum R_j C_j}$$

$$R_B = R_{o1} \parallel R_{o4} \parallel R_{i6}$$

$R_{i6}!!!$

$$\rightarrow \frac{r_{\pi} + R_L}{1 + g_m \cdot r_o}$$



$$R_{o6} = R_{o4c}$$

$$R_{in, ch, chsc} = \frac{2}{g_m}$$

$$R_x = R_{o6} \parallel R_{in, ch, chsc} \approx R_{in, ch} = \frac{2}{g_m}$$

$$[z_A = R_A \cdot C_{pA}]$$

$$[z_B = R_B \cdot C_{pB}]$$

$$[z_D = R_D \cdot C_D]$$

$\uparrow \quad \uparrow$
 $R_{out} \cdot C_L$

$$z_C = R_C \cdot C_{pC}$$

$$\omega_{-3dB} \approx \frac{1}{R_{out} \cdot C_L}$$

$\uparrow$ Common-Mode Range

o/p range ; SR ; OR ; Gain Bandwidth ; $GB = A_d \cdot \omega_{-3dB}$; A_d ; $V_{in, ch}$; P ; power

1) $SR = \frac{I_3}{C_L} \Rightarrow \underline{\underline{I_3 = SR \cdot C_L}}$

slew rate is given by the client

I_4 & I_5 should be larger than I_3 . let $I_4, I_5 = (1.2 \text{ to } 1.5) I_3$
now we have found I_3, I_4, I_5

2) $I_4, I_5 = [1.2 \div 1.5] I_3$

3) $V_{OUT_MAX} = V_{DD} - 2|V_{ovp}| \Rightarrow$

$|V_{ovp}| = \frac{V_{DD} - V_{OUT_MAX}}{2}$

$I = \frac{\beta}{2} \frac{W}{L} (V_{ov})^2$

$[V_{ov}, I] \Rightarrow$

$\frac{W}{L}_4 = \frac{W}{L}_5 = \frac{2 I_4}{\beta'_p \cdot V_{ovp}^2}$

If we know overdrive and current we can find W/L Ratio

$V_{ovG,7} = |V_{ovp}|$

$I_6 = I_7 = ?$

$\frac{W}{L}_6 = \frac{W}{L}_7 = \frac{2 I_6}{\beta'_p \cdot V_{ovp}^2}$

$I_6 = I_4 - \frac{I_3}{2} = I_7$

To compute W/L of M_6 & M_7 we need I_6 & I_7

For nominal operation the current flows equally in two branches of the Differential pair.

4) $\frac{W}{L}_{CM} = \frac{W}{L}_{8,9,10,11}$

Current Mirror Settings

$V_{OUT_MIN} = 2 V_{ovCM} + V_T$

$V_{ovCM} = \frac{V_{OUT_MIN} - V_T}{2}$

$V_{IN_CM} = V_{ov3} + V_{ov4} + V_T$

$(I_{CM}, V_{ovCM}) \Rightarrow \frac{W}{L}_{CM}$

To calculate V_{ov3}

5) $GB = [A_d \cdot \omega_{-3dB}] \left[\frac{\text{rad}}{\text{sec}} \right]$

$A_{d0} = g_{m1} \cdot R_{out}$

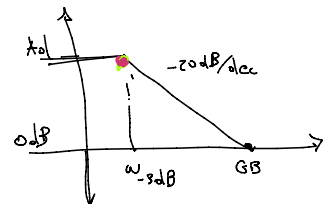
$\omega_{-3dB} = \frac{1}{R_{out} \cdot C_L}$

$GB = \frac{g_{m1}}{C_L}$

$\Rightarrow (g_{m1}) = GB \cdot C_L = \sqrt{2 \beta'_n \frac{W}{L} \cdot I_1}$

$\frac{W}{L}_{1,2} = \frac{GB^2 \cdot C_L^2}{2 \beta'_n \frac{I_3}{2}}$

$I_1 = I_2 = \frac{I_3}{2}$



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$$SR = \frac{I_3}{C_L}$$

$$A_{v0} = g_{m1} \cdot R_{out} \propto g_{m1} \cdot v_o^2 \propto \frac{1}{\sqrt{I}}$$

$$GB = \frac{g_m}{C_L} \propto \sqrt{I}$$

$$P_{diss} = V_{DD} (\sum I)$$

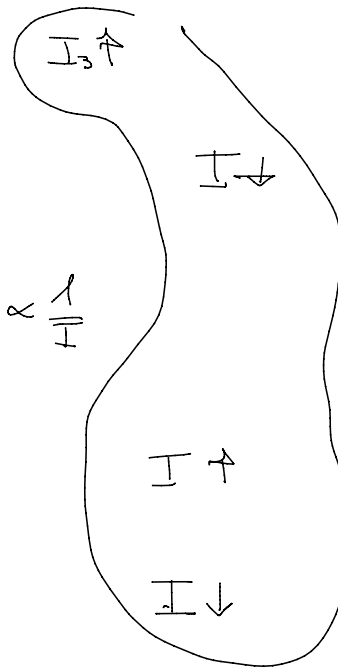
SR ↑

$A_v \uparrow$

$$A_v \propto \frac{1}{I}$$

GB ↑

$P_{diss} \downarrow$



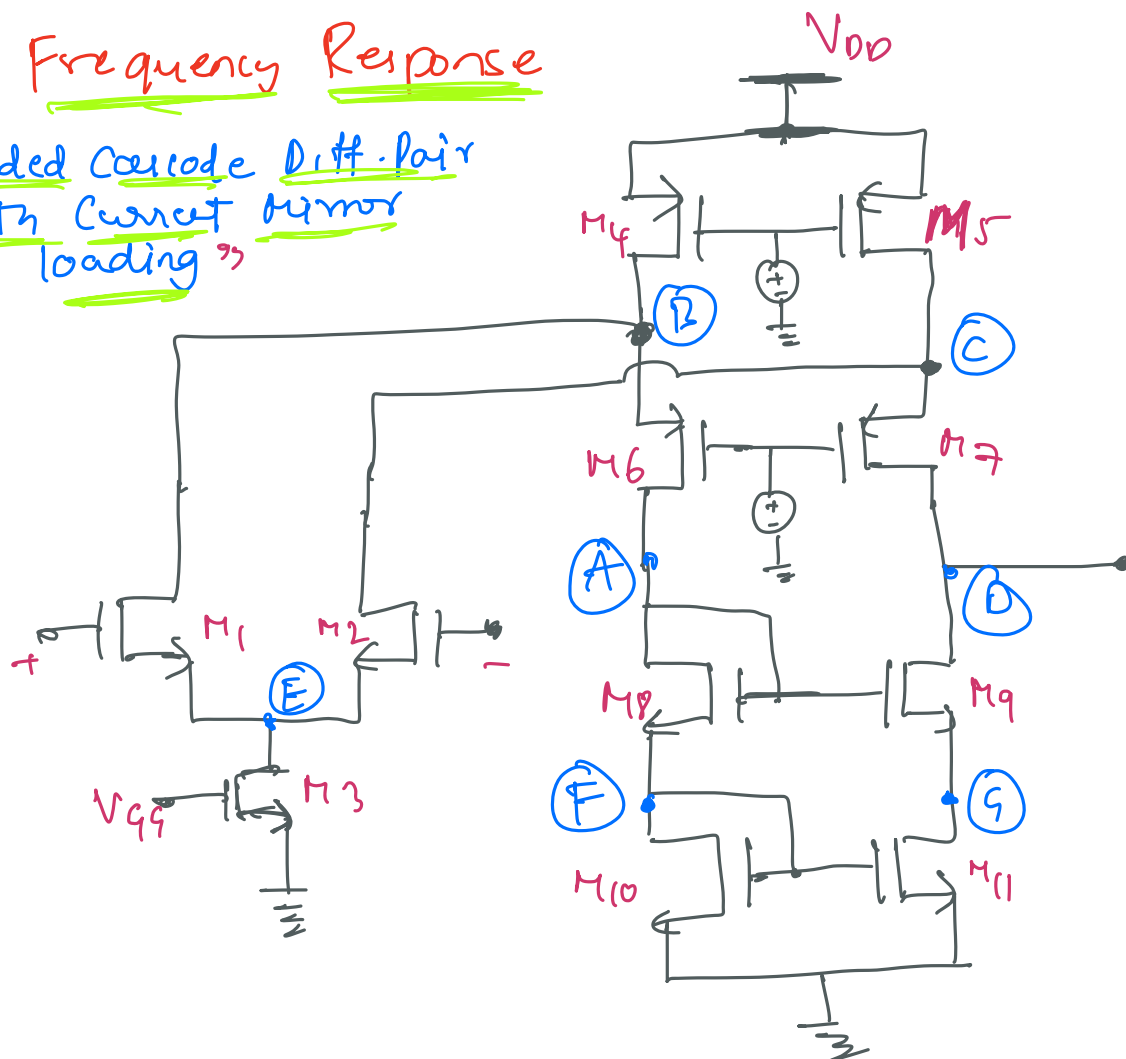
$$g_o = \lambda I_d$$

$$v_o \propto \frac{1}{I_d}$$



Frequency Response

ee Folded Cascode Diff. Pair
with Current Mirror
loading



→ This is a quite complex circuit and it is not

feasible to compute the TF which includes all the parasitic effects to estimate the Dominant Pole in the circuit.

→ estimation of the Dominant pole frequency is calculated using

Zero value Time constant method.

i.e. $\omega_{-3dB} \approx \frac{1}{\sum R_j C_j}$

where

$R_j C_j =$ Time constant of the j^{th} pole

→ So we have to investigate all the nodes basically compute capacitances in parallel with the resistances associated with them individually.

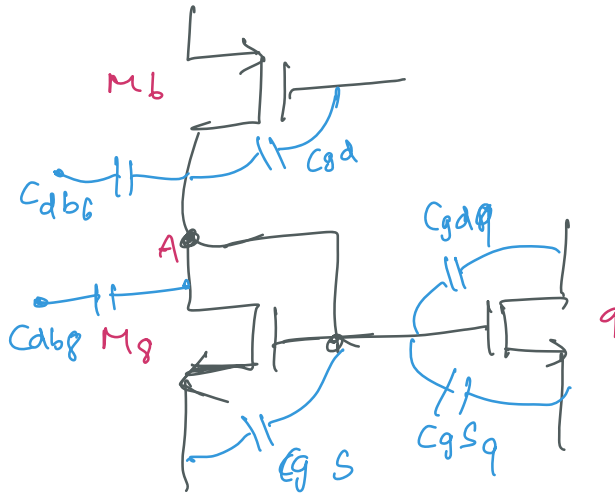
→ let's start with principle frequency.

Note:- node (E) can be ignored because it's virtual Ground in Differential condition.

→ Node A

Let's investigate parasitic resistances associated with Node A

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For Node A

Resistance at Node A = $R_{o6} \parallel R_{in,CM,casc}$
to the Ground

output resistance of M_6

Input resistance of cascode CM

R_{o6} = It is resistance seen from Node A upwards.

= output resistance of the Folded cascode.
(It is quite large.)

$$R_{IN_{CM_{casc}}} = \frac{2}{g_m} =$$

Input Resistance of the cascode current mirror which is nothing but two Diode connected transistors in series

Between R_{os} & $R_{IN_{CM_{casc}}}$ the later is very less and

$$R_A = R_{os} \parallel R_{IN_{CM_{casc}}} \Rightarrow R_{IN_{CM_{casc}}} \Rightarrow \frac{2}{g_m}$$

$\therefore$ The R_A is very less the Time constant associated with the node is also less

$\therefore$

$$\tau_A = R_A C_{pA}$$

Capacitance associated with node.

$\therefore$ we can neglect the Node A

→ For Node B

Let's First look at the Resistance at Node $\textcircled{B}$

$$R_B = R_{O1} \parallel R_{O4} \parallel R_{i6}$$

output
Resistance of
Transistor 1 & 4

i/p Resistance of
 M_6

As M_6 is a CG stage
the input Resistance is
almost negligible

$\therefore R_B$ is also much less and thereby
the Time constant of the Node is also
very less

Now the effect of Node B can
also be neglected.

R_{i6} is very small

$$\approx \frac{r_{o6} \parallel R_L}{1 + g_m r_o}$$

Node C

node C has similar situation as Node B

$\therefore$ effects of Node C can be ignored as time constant associated with this node is very small.

Node D

Z_D is by far greatest contribution to the Time constant. Total time constant becomes

both R_D & C_D are very large.

It is the output resistance is very large

It has effects of C_L and parasitic capacitances of the load.

Node F & G

The effects can be neglected because they are low impedance nodes.

node E

node E in differential configuration is a virtual Ground.

$\omega_{3dB} \approx \frac{1}{Rout \cdot CL}$

By

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